

Proof Effort Summary

ShuYu

1 Problem Overview

Problem (Remark 7.22, Matsui–Takahashi 2017). Let R be a commutative noetherian ring and $x \in R$ a nonzerodivisor. For a positive integer d and non-negative integers $a_0, \dots, a_d$, define

$$X(a_0, a_1, \dots, a_d) = \bigoplus_{i \geq 0} \left(R/x^{a_0 i^d + a_1 i^{d-1} + \dots + a_d} \right) [i].$$

Does the equality

$$\text{thick}^\otimes \{X(a_0, \dots, a_d) \mid a_0, \dots, a_d \geq 0\} = \text{thick}^\otimes \{X(1, 0, \dots, 0)\}$$

hold for all $d \geq 3$? The case $d = 2$ was proven in Example 7.21 of the same paper.

Classification.

- **Area:** Commutative algebra / homological algebra; thick tensor ideals in $D^-(R)$ and tensor triangular geometry.
- **Type:** Equality of thick tensor ideals. The non-trivial direction is showing every generator on the left belongs to the single-generator ideal on the right.
- **Difficulty:** Hard; an explicitly stated open research problem, unresolved for about nine years. The $d = 2$ proof exploits a specific algebraic coincidence, namely 2-fold even/odd splitting via Corollary 7.3, which does not directly generalize because $d!$ has odd prime factors for $d \geq 3$.

2 Final Proof Status

Result: PASS. A correct proof was found.

The proof establishes the affirmative answer for all $d \geq 1$, fully resolving Remark 7.22. The key strategy is:

1. **Generalize 2-fold splitting to p -fold splitting (Lemma B):** for any $p \geq 2$, a complex $\mathbf{X}(a_i)$ belongs to the thick tensor ideal generated by its p -fold sub-sampled parts $\mathbf{X}(a_{pi+r})$, for $r = 0, \dots, p-1$. This overcomes the fundamental obstruction of dividing by odd primes.
2. **Bottom-up induction on monomial degree:** starting from $\mathbf{X}(i^d) \in \mathcal{T}$, iterate finite differences (Lemma D) to produce $\mathbf{X}(\Delta^m(i^d)) \in \mathcal{T}$, subtract lower-order terms (Lemma C) using the inductive hypothesis, and apply c -fold splitting with $c = d!/k!$ to recover $\mathbf{X}(i^k)$.

The proof was verified with 78 out of 79 miniclaims passing, zero failures, and one UNCERTAIN point: a minor expository gap regarding the trivial case $k = d$ in the c -fold splitting step. This is mathematically inconsequential since $\mathbf{X}(i^d)$ is already the generator of $\mathcal{T}$.

3 Round-by-Round Summary

3.1 Round 1 (Exploration)

All three proof search agents, Claude, Codex, and Gemini, produced empty proofs with only placeholder text. However, the verification phase was still useful: Claude’s verification report identified the fundamental algebraic obstruction. For $d \geq 3$, the finite difference cascade produces leading coefficients divisible by $d!$, which contains odd prime factors, while Corollary 7.3 of the paper only provides 2-fold even/odd splitting, that is, division by 2. This diagnosis shaped the subsequent strategy.

Verdict. FAIL. No proof content was produced.

Key output. Identification of the odd-prime obstruction.

3.2 Round 2 (Breakthrough)

Claude produced a complete, mathematically rigorous proof generalizing the result to all $d \geq 1$. The proof introduced four new lemmas, labeled A–D, and used bottom-up induction on monomial degree. The central innovation was Lemma B, the p -fold splitting lemma, which generalizes Corollary 7.3 from $p = 2$ to arbitrary $p \geq 2$. Verification confirmed 22 miniclaims, all PASS, before timing out. Codex and Gemini again produced empty proofs.

Verdict. FAIL, because verification was incomplete due to timeout.

Key output. The correct proof strategy was found.

3.3 Round 3 (Refinement)

All three models adopted Claude’s proof strategy from Round 2. Gemini’s version was selected because it had partial verification data, with 5 out of 30 miniclaims verified and all passing. The proofs added clarifications about $Y_r \in D^-(R)$ and the K -flatness of free modules. Verification again timed out before the core mathematical content could be fully checked.

Verdict. FAIL, because verification was incomplete.

Key output. The proof exposition improved, but verification remained the bottleneck.

3.4 Round 4 (Verification Success)

The proof was further refined with explicit labeled properties (T1)–(T5) of thick tensor ideals, detailed uniqueness proofs for the diagonal extraction in Lemma B, explicit well-definedness and injectivity checks in Lemma C, and a computational verification section. Both Claude and Codex received PASS verdicts with 78/79 miniclaims passing. Claude was selected as the final proof.

Verdict. PASS.

Key output. Fully verified proof.

4 Approaches Tried

#	Strategy	Rounds	Outcome
1	No proof attempted; empty placeholders only	R1 (all models), R2 (Codex, Gemini)	Abandoned. The models failed to produce content in the early rounds.
2	p -fold splitting + finite differences + bottom-up induction	R2–R4 (Claude, then all models)	Carried forward and verified. This was the winning strategy.

Only one substantive proof strategy was attempted across all rounds. The key evolution was not in the mathematical approach but in the level of exposition: each round added more explicit justifications to help the verification agents complete their checks within the time budget.

Specific refinements across rounds.

- **R2 to R3:** Added explicit justification for $Y_r \in D^-(R)$ and K -flatness arguments for the derived tensor product.
- **R3 to R4:** Added labeled properties (T1)–(T5), a uniqueness proof for diagonal extraction, explicit injectivity verification, an iterated Lemma D proof, and computational verification for small cases.

5 Key Mathematical Insights

5.1 Core Discovery: p -fold Splitting (Lemma B)

The central insight that unlocks the general case is that Corollary 7.3’s even/odd decomposition generalizes to arbitrary p -fold splitting. Given a complex

$$X = \bigoplus_{n \geq 0} M_n[n]$$

with zero differentials, define

$$X^{(r)} = \bigoplus_{i \geq 0} M_{pi+r}[i]$$

for $r = 0, \dots, p-1$. Then

$$X \in \text{thick}^{\otimes} \{X^{(0)}, \dots, X^{(p-1)}\}.$$

The proof constructs an auxiliary complex

$$Y_r = \bigoplus_{j \geq 0} R[(p-1)j + r],$$

tensors it with $X^{(r)}$, and extracts the diagonal direct summand

$$Z_r = \bigoplus_{i \geq 0} M_{pi+r}[pi + r].$$

This is the tool that enables division by any integer p , overcoming the odd-prime obstruction.

5.2 Coefficient Formula via Stirling Numbers (Lemma A)

The m -th forward finite difference of i^d has the explicit expansion

$$\Delta^m(i^d) = \sum_{j=0}^{d-m} \binom{d}{j} m! S(d-j, m) i^j,$$

where all coefficients are non-negative integers. This non-negativity is essential: it ensures that the lower-order remainder

$$q(i) = \Delta^m(i^d) - \frac{d!}{k!} i^k$$

has non-negative coefficients, enabling subtraction via Lemma C.

5.3 Induction Architecture

The proof proceeds by induction on k from 0 to d , showing $\mathbf{X}(i^k) \in \mathcal{T}$.

- **Base case** ($k = 0$): Extract R/x as a direct summand of $\mathbf{X}(i^d)$, then tensor with $\bigoplus_{i \geq 0} R[i]$.
- **Inductive step** ($k \geq 1$): Apply Δ^{d-k} to i^d via Lemma D, subtract lower-order terms using Lemma C and the inductive hypothesis, and use c -fold splitting with $c = d!/k!$ via Lemma B to recover $\mathbf{X}(i^k)$.

5.4 Structural Observation

The proof reveals that the case $d = 2$ is not genuinely special. The same mechanism works uniformly for all d . The apparent obstruction at odd primes was an artifact of the limited toolkit, namely only having 2-fold splitting, rather than a genuine mathematical barrier.